9 Vector Fields

Smoothness, Lie Brackets & Integral Curves

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Vector Fields

Smooth curves on a closed interval

Definition (Smooth curve)

A mapping $\sigma : [a, b] \to M$ is a *smooth curve* in M if σ extends to a C^∞-map of $(a - \varepsilon, b + \varepsilon) \to M$ for some $\varepsilon > 0$.

Note

If $\sigma : [a, b] \to M$ is a smooth curve, then its tangent vector

$$\dot{\sigma}(t) = \sigma_* \left(\frac{d}{dt}\Big|_t \right) \in T_{\sigma(t)} M$$

is well-defined for all $t \in [a, b]$.

Vector fields along a curve

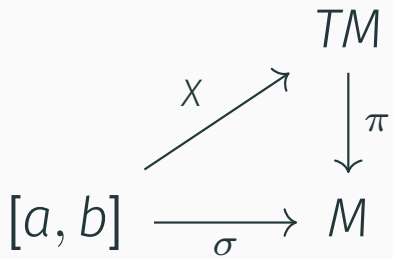
Definition (Vector field along a curve)

A smooth vector field X along a curve $\sigma : [a, b] \to M$ is a C^∞-map

$$X : [a, b] \longrightarrow TM$$

which *lifts* σ, i.e.

$$\pi \circ X = \sigma.$$



Vector fields on an open subset

Definition (Vector fields on an open subset of M)

A smooth vector field on an open set U in M is a C^∞-map $X : U \to TM$ s.t.

$$\pi \circ X = \mathrm{id}_U.$$

Sometimes we call a smooth vector field X on M a *section* of TM. The set of all smooth sections on U of TM is denoted by $\Gamma(U; TM)$. It is a vector space over $\mathbb{R}$ and also a $C^\infty(U)$-*module*:

$$\Gamma(U; TM) = C^\infty(U, TM) = \mathfrak{X}(U).$$

A Criterion for Smoothness

Three equivalent conditions

Proposition

Let X be a vector field on M. Then the following are equivalent:

(a) *X is C^∞;*

(b) *if $(U, x_1, \ldots, x_d)$ is a local coordinate system on M and $\{a_i\}$ are the functions on U defined by*

$$X|_U = \sum_{i=1}^{d} a_i \frac{\partial}{\partial x_i},$$

then $a_i \in C^\infty(U)$;

(c) *whenever $V \subset M$ is open, if $f \in C^\infty(V)$, then $X(f) \in C^\infty(V)$.*

Proof: (a) $\Rightarrow$ (b)

Let $X : M \to TM$ be a C^∞-map s.t. $\pi \circ X = \mathrm{id}_M$, so $X(p) = V_p \in T_pM$. Take a local coordinate system $(U, \varphi, x_1, \ldots, x_d)$ about $p \in M$; the local coordinate system about V_p is given by the local trivialization $\widetilde{\varphi} : \pi^{-1}(U) \to \mathbb{R}^{2d}$.

Smoothness of X in local coordinates requires $\widetilde{\varphi} \circ X \circ \varphi^{-1}$ to be C^∞ from $\varphi(U) \subset \mathbb{R}^d$ to $\widetilde{\varphi}\big(\pi^{-1}(U)\big) \subset \mathbb{R}^{2d}$. Writing $X(p) = V_p = \sum_i a_i(p) \left.\frac{\partial}{\partial x_i}\right|_p$,

$$\begin{aligned}\widetilde{\varphi} \circ X \circ \varphi^{-1}(x_1, \ldots, x_d) &= \Big(\varphi\big(\varphi^{-1}(x_1, \ldots, x_d)\big);\ a_i\big(\varphi^{-1}(x_1, \ldots, x_d)\big)\Big) \\ &= \big(x_1, \ldots, x_d;\ a_i(\varphi^{-1}(x_1, \ldots, x_d))\big),\end{aligned}$$

which means $a_i \in C^\infty(U)$.

Proof: (b) $\Rightarrow$ (c) $\Rightarrow$ (a)

(b) $\Rightarrow$ (c). For $f \in C^\infty(V)$ we show $X(f)$ is C^∞ about each $p \in V$. Take a local coordinate system $(U, x_1, \ldots, x_d)$ about p; then $X\big|_U = \sum_i a_i \frac{\partial}{\partial x_i}$ with $a_i \in C^\infty(U)$, so

$$X(f) = \sum_{i=1}^{d} a_i \frac{\partial f}{\partial x_i} = \sum_{i=1}^{d} a_i \frac{\partial (f \circ \varphi^{-1})}{\partial x_i} \in C^\infty(U).$$

(c) $\Rightarrow$ (a). In an arbitrary local coordinate system $(U, x_1, \ldots, x_d)$, write $X = \sum_i a_i \frac{\partial}{\partial x_i}$. Apply (c) to the coordinate function $x_i \in C^\infty(U)$:

$$X(x_i) = a_i \in C^\infty(U).$$

The Lie Bracket

The Lie bracket of two vector fields

If X, Y are smooth vector fields on M (i.e. $X, Y \in \mathfrak{X}(M) = \Gamma(TM)$), we define a vector field $[X, Y]$, the *Lie bracket* of X and Y, by

$$[X, Y]_p(f) = X_p\big(Y(f)\big) - Y_p\big(X(f)\big).$$

Note

The bracket measures the *failure* of the flows of X and Y to commute; locally it is the commutator of the derivations X and Y.

Properties of the Lie bracket

Proposition

Let $X, Y, Z \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$.

(a) $[X, Y]$ *is a smooth vector field on* M;

(b) $[fX, gY] = fg\,[X, Y] + f\,(Xg)\,Y - g\,(Yf)\,X$;

(c) $[X, Y] = -[Y, X]$ *(skew-symmetry);*

(d) $\big[[X, Y], Z\big] + \big[[Y, Z], X\big] + \big[[Z, X], Y\big] = 0$ *(Jacobi identity).*

Note (Lie algebra)

A vector space with a bilinear operation satisfying (c) and (d) is called a *Lie algebra*.

Integral Curves

Integral curves

Definition (Integral curve)

Let $X \in \mathfrak{X}(M)$. A smooth curve σ in M is an *integral curve* of X if

$$\dot{\sigma}(t) = X\big(\sigma(t)\big) \qquad (\forall\, t \in I).$$

Reformulation

A curve $\gamma : (a, b) \to M$ is an integral curve of $X \iff$

$$\gamma_*\Big(\frac{d}{dt}\Big|_t\Big) = X\big(\gamma(t)\big) \qquad (t \in (a, b)).$$

Existence and uniqueness: local form

Suppose $0 \in (a, b)$ and $\gamma(0) = p$. Choose a local coordinate system (U, φ) with coordinate functions $x_1, \ldots, x_d$ about p; write

$$X|_U = \sum_{i=1}^{d} f_i \frac{\partial}{\partial x_i} \qquad (f_i \in C^\infty(U)).$$

For all t with $\gamma(t) \in U$,

$$\gamma_*\left(\frac{d}{dt}\Big|_t\right) = \sum_{i=1}^{d} \frac{d(x_i \circ \gamma)}{dt}\Big|_t \left.\frac{\partial}{\partial x_i}\right|_{\gamma(t)} = X(\gamma(t)) = \sum_{i=1}^{d} f_i(\gamma(t)) \left.\frac{\partial}{\partial x_i}\right|_{\gamma(t)}.$$

Reduction to a system of ODEs

Thus γ is an integral curve of X on $\gamma^{-1}(U)$ iff, with $\gamma_i \triangleq x_i \circ \gamma$,

$$(\star) \qquad \left.\frac{d\gamma_i}{dt}\right|_t = f_i\big(\varphi^{-1}\big(\gamma_1(t), \ldots, \gamma_d(t)\big)\big) \qquad (t \in \gamma^{-1}(U)).$$

Theorem (Existence & uniqueness)

($\star$) is a system of ODEs, for which there exists a fundamental existence & uniqueness theorem; hence integral curves of a smooth vector field exist and are unique.